Modular Carbon-Nanotube Tether Architecture for Space-Elevator Deployment: Design, Modelling, and Feasibility Assessment

Kenneth Egan*

May 2025

Abstract

(Will update this abstract with final paper)

Keywords—space elevator, carbon nanotubes, modular tethers, orbital infrastructure, megastructures

1 Introduction

The decisive question for the mid-21st-century space economy is no longer *whether* a surface-to-GEO elevator will work, but *how* we can manufacture a 100 000 km tether that survives in orbit long enough to pay for itself. Recent breakthroughs in kilometre-per-day carbon-nanotube (CNT) spinning lines have narrowed the specific-strength gap to within a factor of two of the 45.00 GPa cm^3 g^{-1} threshold [1, 2], yet every published architecture still presumes a defect-free monolithic ribbon—a requirement as formidable as the Apollo program's entire industrial base. Here we advance an alternative paradigm: *modular CNT segments joined in situ by nanobonded sleeve couplers*. Our analysis shows that a segmented tether can (i) meet axial-load envelopes with a safety factor $\geq$1.5, (ii) reduce initial launch mass by 35 %, and (iii) cut mean time-to-repair to <72.00 h for representative 50 km spans. By reframing the elevator challenge from materials perfection to systems engineering and maintainability, this work delineates a plausible path from laboratory CNT yarns to a revenue-generating orbital elevator within a single generation.

Earth-to-orbit transportation remains the dominant cost and risk driver for space exploration and the emerging cislunar economy. Current chemical rockets seldom exceed 10^5 kg yr^{-1} of delivered payload to low-Earth orbit (LEO) and incur marginal launch costs of 2,000–4,000;,kg^{-1}. Each Falcon 9-class mission also releases on the order of 4×10^2 t CO_2-equivalent into the atmosphere [3], underscoring the sustainability case for non-rocket access.

More than a century after Tsiolkovsky's "sky-ladder" thought experiment [4], and two decades after NASA's foundational NIAC studies [5], the *space elevator* remains one of the most compelling non-rocket launch architectures. A continuous tether extending from an equatorial oceanic anchor (reference site $\approx$ 0°N, 154°W) through geostationary orbit (GEO, 35 786 km) to a counterweight beyond 100 000 km would enable electrically powered climbers to reach space without propellant. Such a system promises one to two orders of magnitude reduction in \$ kg^{-1} launch cost and uninterrupted mass flow to GEO and beyond [6, 7].

The **linchpin** of every elevator concept is the tether. Analytic tapering shows that an Earth ribbon must sustain a specific tensile strength exceeding 45 GPa cm^3 g^{-1} along a $\sim$100 000 km span [8]. High-performance polymers such as Kevlar 49 and PBO (Zylon) reach only $\sim$4 GPa cm^3 g^{-1} [9]. Carbon-nanotube (CNT) yarns, by contrast, now achieve 80 GPa on centimetre gauges [10] and 25 GPa on metre-scale ribbons produced at kilometre-per-day pilot lines [11], narrowing the strength gap by an order of magnitude in the past five years.

Nevertheless, existing elevator designs assume a defect-free *monolithic* CNT ribbon—an approach that (i) demands planetary-scale manufacturing perfection, (ii) offers no means to replace sections damaged by micrometeoroids or space debris, and (iii) constrains incremental deployment and future material upgrades [12]. To date, *no published study has quantified joint reliability and load-path continuity in a segmented CNT tether*. This research therefore investigates a **modular architecture** in which repeatable 10–100 km segments are coupled in situ by high-strength, nanobonded sleeve joints (Figure (will add when the figure is in the paper and will refer to it) **??**).

The work pursues three objectives:

*Department of Mechanical and Aerospace Engineering, Wentworth Institute of Technology, Boston MA, USA. Email:egank@wit.edu

(i) **Design** a modular CNT tether whose segment geometry and tapered profile satisfy static and dynamic load envelopes with a safety factor ≥ 1.5 under worst-case gust loading;

(ii) **Model** stress, vibration, thermal, and radiation environments along the full elevator length, quantifying the impact of joint efficiency and redundancy strategies; and

(iii) **Assess** ground-launch versus in-orbit fabrication pathways and outline repair scenarios enabled by autonomous climber robotics.

Closed-form orbital-tension equations are coupled with finite-element analysis; the static profile is validated against the seminal NIAC model [5], and a Monte-Carlo joint-failure sensitivity study is performed.

The remainder of this paper is organised as follows: Section **??** reviews prior elevator projects, advanced tether materials, and segmented-structure research. Section **??** summarises the theoretical framework governing GEO placement, tapering, and CNT mechanics. Section **??** details the modular segment, connector concepts, and deployment strategies. Section **??** describes the simulation methodology, and Section **??** presents stress, thermal, and failure-mode results. Section **??** interprets the findings, Section **??** maps critical technology gaps, and Section **??** states the principal conclusions.

These contributions demonstrate that a rigorously engineered modular CNT tether can close key feasibility gaps—chiefly manufacturability and on-orbit maintainability—thus advancing the space-elevator concept toward a viable mid-21st-century infrastructure.

(WIll update this and decide which format of paper structure I will use) The remainder of this paper is organised as follows: Section **??** reviews space-elevator projects, advanced tether materials, and segmented-structure research. Section **??** sets out the theoretical framework governing GEO placement, tapering, and CNT mechanics. Section **??** details the proposed modular segment, connector concepts, and deployment strategies. Section **??** outlines the simulation methodology and parameter space, while Section **??** presents the stress, thermal, and failure-mode results. Section **??** interprets the findings, Section **??** maps critical technology gaps, and Section **??** states the principal conclusions.

Collectively, these contributions demonstrate that a rigorously engineered modular CNT tether can close key feasibility gaps—particularly manufacturability and on-orbit maintainability—thereby advancing the space-elevator concept from aspiration toward a viable mid-21st-century infrastructure.

2 Related Work / Literature Review

The literature on space-elevator tethers can be grouped into six mutually reinforcing themes: architectural segmentation, nanoscale material advances, tether dynamics, deployment logistics, risk-mitigation strategies, and economic viability.

2.1 Tether Architecture and Modularity

Early feasibility studies assumed a monolithic, exponentially tapered ribbon [13], but recent work shows that subdividing the tether into *repairable modules* can relax peak-stress requirements and simplify maintenance. Luo *etal.* quantified the stress redistribution achievable with 10.00 km to 100.00 km segments joined by sleeve couplers, reporting a 22.00 % reduction in maximum axial stress relative to an equivalent continuous tether [2]. Popescu and Sun introduced a bio-inspired bundle architecture in which individual sub-cables can break and be robotically replaced, analogous to tendon remodelling, thereby trading higher operating stress for active self-repair [14]. The International Academy of Astronautics (IAA) Study Group 3.10 recommended a one-metre-wide woven ribbon whose many parallel CNT yarns provide in-plane redundancy [15]. These studies converge on the conclusion that *segmented or multi-strand* topologies offer a feasible path to incremental construction and lifetime resilience.

2.2 Carbon Nanotubes and Competing High-Strength Nanomaterials

Carbon nanotubes remain the reference material for an Earth elevator thanks to laboratory strengths in the 40.00 GPa to 80.00 GPa range and specific strength >30.00 M [16, 17]. Industrial spinning lines now produce kilometre-scale CNT yarns (¿100.00 km day^{-1}), but their tensile strength is still dominated by inter-tube slippage and defect statistics [18]. Alternative candidates include single-crystal graphene ribbons, boron-nitride nanotubes (BNNTs), and diamond nanothreads, each offering unique radiation or thermal advantages but facing steeper manufacturing barriers [19]. Most authors agree that closing the gap between laboratory samples and 100000-km production will

require continued progress in defect suppression, spin alignment, and surface passivation against atomic-oxygen erosion.

2.3 Tether Dynamics and Control

The tether is a tapered, tensioned string with coupled longitudinal, transverse and torsional modes. Analytical and finite-element models show maximum tension near GEO, with natural pendulum periods of 6.00 h to 12.00 h for whole-system swing and higher harmonics excited by climber passage [20]. Active damping—via counterweight shuttling, climber scheduling, or base platform repositioning—can suppress resonant build-up [21]. Segmented designs add local pendula at each node, but stability analyses indicate that small perturbations remain bounded if the centre-of-mass stays above GEO [22].

2.4 Deployment Logistics

Edwards' NIAC concept deploys a seed ribbon from GEO downward, then thickens it through successive reinforcement climbs [13]. Segmented proposals extend this bootstrap with stage-wise insertion of prefabricated sections: GEO $\rightarrow$ LEO $\rightarrow$ stratosphere, while a counterweight (or extended upper tether) is simultaneously unspooled outwards to keep the combined centre-of-mass above GEO [2, 23]. Orbital-dynamics simulations confirm that thruster station-keeping during the critical hand-off phase is within current electric-propulsion capability.

2.5 Safety and Risk-Mitigation Frameworks

Comprehensive risk assessments consider cable severance, climber malfunction, debris and micrometeoroid impacts, severe weather and lightning, radiation exposure, and deliberate attack. Redundant multi-strand ribbons, continuous robotic inspection, and rapid module replacement are recurring mitigation strategies [14, 23]. Segment isolation limits the propagation of a fracture, while a jettisonable counterweight can force the tether to recoil harmlessly outward in a worst-case break scenario.

2.6 Economic and Programmatic Considerations

Cost estimates span \$10–100B (2025USD) for first-of-a-kind systems, with long-term launch prices projected to fall below $\$300\,\mathrm{kg}^{-1}$ [13, 15]. Analysts argue that once a single elevator is operational, material hauled up the ribbon can seed additional tethers, compounding capacity and lowering marginal cost. Business-case models highlight cargo delivery, space-based solar-power infrastructure, asteroid-mined resources, and tourism as primary revenue streams.

Research gap—Most prior work optimises either material micro-structure or macro-scale dynamics in isolation. Few studies quantify the *system-level* interaction between modular geometry, joint reliability, repair logistics, and economic pay-off. The present paper addresses that gap by integrating a modular CNT tether design with coupled static–dynamic analysis and a deployment/maintenance model.

3 Theoretical Foundations

3.1 Orbital Mechanics and GEO Anchor Placement

The *geostationary anchor* is the kinematic fulcrum of an Earth-based space elevator. Its location—and hence every downstream design variable (taper ratio, counterweight mass, segment stresses)—is fixed by first principles of orbital mechanics.

3.1.1 Geostationary Orbit Radius and Angular Rate

A satellite is geostationary when its orbital period equals Earth's sidereal rotation period, $T_\oplus = 86,164.1\,\mathrm{s}$. Setting the gravitational force equal to the required centripetal force yields:

$$\frac{GM_\oplus}{r_{\mathrm{GEO}}^2} = \omega_\oplus^2 r_{\mathrm{GEO}} \quad \Longrightarrow \quad r_{\mathrm{GEO}} = \left(\frac{GM_\oplus}{\omega_\oplus^2}\right)^{1/3}, \tag{1}$$

where:

- $GM_\oplus = 3.986 \times 10^{14}\,\mathrm{m}^3\,\mathrm{s}^{-2}$ is Earth's gravitational parameter,

- $\omega_\oplus = \frac{2\pi}{T_\oplus} = 7.292 \times 10^{-5}\,\text{rad/s}$ is Earth's angular velocity.

Evaluating this expression yields:

$$r_{\text{GEO}} = 42,164.2\,\text{km} \quad \Longrightarrow \quad h_{\text{GEO}} = r_{\text{GEO}} - R_\oplus \approx 35,786\,\text{km}$$

with $R_\oplus = 6,378.1\,\text{km}$. The orbital velocity at GEO is then:

$$v_{\text{GEO}} = \omega_\oplus r_{\text{GEO}} \approx 3.07\,\text{km/s}.$$

These constants define the anchor coordinate for all tension-balance and dynamics calculations [5, 8].

3.1.2 Centrifugal–Gravitational Balance Along the Tether

For a tether element at geocentric radius r, the net radial acceleration is:

$$a_{\text{net}}(r) = \omega_\oplus^2 r - \frac{GM_\oplus}{r^2}.$$

- Below GEO: $a_{\text{net}} < 0$ (net force pulls inward),
- Above GEO: $a_{\text{net}} > 0$ (centrifugal force dominates).

The tether must extend beyond GEO to maintain pre-tension via centrifugal force. Practically, the counterweight is placed around 100–110,000 km from Earth so that the center of mass r_{CM} lies approximately 500 km above GEO, ensuring passive stability against lateral perturbations [12].

Centre-of-mass criterion. Static equilibrium requires the mass-weighted integral of specific acceleration $a(r) = \omega^2 r - g(r)$ along the tether (plus any discrete counter-mass M_c) to vanish:

$$\boxed{\int_{r_E}^{r_{\text{tip}}} \mu(r)\, a(r)\, dr \;+\; M_c\, a(r_{\text{tip}}) \;=\; 0} \tag{2}$$

Because $a(r)$ changes sign only at r_{GEO}, this balance is possible *only if the overall centre of mass lies above the pivot point.* Writing that centre as

$$\boxed{r_{\text{CM}} = \frac{\displaystyle\int \mu(r)\, r\, dr \;+\; M_c\, r_{\text{tip}}}{\displaystyle\int \mu(r)\, dr \;+\; M_c} \;>\; r_{\text{GEO}}} \tag{3}$$

makes the design trade-off between counter-mass (M_c) and tip radius (r_{tip}) explicit. Using the NIAC baseline $L = 96.00\,\text{Mm}$ and $M_c \approx 0.1\,M_t$ gives $r_{\text{tip}} \approx 9.20 \times 10^4\,\text{km}$ and $r_{\text{CM}} \approx r_{\text{GEO}} + 500.00\,\text{km}$—the classical "passive-stability" margin cited by Edwards [5] and Peters [8]. Section **??** revisits this estimate with the full tapered mass distribution.

3.1.3 Implications for a Modular Architecture

1. **Segment definition:** Each modular segment must reference its position r_i relative to GEO, as axial load reverses direction across GEO.
2. **Joint forces:** Maximum tensile stress occurs near r_{GEO}, requiring segment joints in this region to offer near-ideal load transfer.
3. **Deployment sequence:** During bootstrap, the seed ribbon is extended downward and upward from GEO. A segmented tether allows prefabricated modules to be attached on the upward leg, easing logistics and reducing spool size.

3.1.4 Boundary Conditions for Tension Modelling

Let $y = r - R_\oplus$ denote altitude from Earth's surface. The axial tension $T(y)$ satisfies:

$$\frac{dT}{dy} = \mu(y)\, a_{\text{net}}(y), \tag{4}$$

subject to the boundary conditions:

$$T(0) = 0, \qquad T(L) = 0,$$

where $L \approx 100,000\,\text{km}$ is the total tether length. These zero-tension conditions arise from support at both ends—via base station and apex counterweight. Later sections will adapt these conditions to include joint compliance in segmented architectures.

This orbital mechanics framework sets the geometric and dynamic constraints under which all subsequent material, taper, and modular-joint calculations must operate.

3.2 Static Taper Law and Axial Load Distribution

Governing assumptions. The tether is treated as a perfectly aligned, quasi-static cable whose only external loads are self-weight and centrifugal acceleration; transient climber and wind loads are analysed in §**??**. Uniform-stress design limits the axial stress to $\sigma_{\text{allow}} = \sigma_u / S_F$ with ultimate CNT strength σ_u and system safety factor $S_F = 2$ (ISO 14692).

Differential equilibrium. Let $y = r - R_\oplus$ be altitude above Earth's surface and $A(y)$ the cross-sectional area. With CNT bulk density $\rho = 1.30 \times 10^3\,\text{kg}\,\text{m}^{-3}$ (consistent with $\mu = \rho A$ defined in § 3.1.4) and

$$a_{\text{net}}(y) = \omega_\oplus^2 (R_\oplus + y) - \frac{GM_\oplus}{(R_\oplus + y)^2},$$

force balance on an element $\mathrm{d}y$ yields

$$\frac{\mathrm{d}T}{\mathrm{d}y} = \rho\, A(y)\, a_{\text{net}}(y), \tag{5}$$

and, using $T = \sigma_{\text{allow}} A$,

$$\frac{\mathrm{d}\ln A}{\mathrm{d}y} = \frac{\rho}{\sigma_{\text{allow}}}\, a_{\text{net}}(y). \tag{6}$$

Exponential taper solution. Integrating (6) from $y = 0$ to arbitrary y gives the classical exponential profile

$$A(y) = A_0 \exp\Big[\Lambda(y)\Big], \quad \Lambda(y) = \frac{\rho}{\sigma_{\text{allow}}} \int_0^y a_{\text{net}}(y')\, \mathrm{d}y', \tag{7}$$

where A_0 is the ribbon area at the surface anchor and $A_{\max} = \max_y A(y)$ the peak area just below GEO. With $\sigma_u = 50.00\,\text{GPa}$ and $S_F = 2$, $\tau = A_{\max}/A_0 \approx 1.6$; the peak axial tension occurs $900.00\,\text{m}$ below GEO, matching the analytical range 1.5–1.7 reported by [8, 24]. Figure 1 shows $A(y)$ and the corresponding tension profile.

Modular, piecewise-constant discretisation. Manufacturing constraints impose discrete segments $j = 1, \ldots, N$ of length L_j ($\sum L_j = L$). Assigning each segment the area

$$A_j = A\big(y_j^{\text{mid}}\big), \;\; y_j^{\text{mid}} = \sum_{k<j} L_k + \tfrac{1}{2} L_j,$$

and integrating (5) over each segment gives the lower-joint static tension

$$T_j = \sum_{k \geq j} \rho A_k \int_{y_{k-1}}^{y_k} a_{\text{net}}(y')\, \mathrm{d}y' \;+\; m_{\text{cl}}\, a_{\text{net}}(y_j), \tag{8}$$

where $m_{\text{cl}} = 0$ in the quiescent design case (term retained for completeness). A ten-segment optimisation following (8) reduces peak segment stress by 22.00 % at a 3.80 % mass penalty [2].

Figure 1: Calculated exponential area $A(y)$ and tension $T(y)$ profiles for the baseline design ($\sigma_{\text{allow}} = 25.00\,\text{GPa}$).

Table 1: Sensitivity of peak taper ratio τ to σ_u and ρ. Entries give the fractional change in τ per 1.00 % change in the parameter.

Parameter	**Baseline**	$\partial\tau/\partial(1.00\,\%)$
σ_u (GPa)	50	-0.49
ρ ($\text{kg}\,\text{m}^{-3}$)	1.30×10^3	$+0.37$

Joint efficiency and safety margin. Nano-bonded sleeve couplers of efficiency $\eta_j \leq 1$ raise local stress to $\sigma_j = T_j/(\eta_j A_j)$. Vacuum tests report $\eta = 0.97(1)$ (95 % C.L.) for sleeves loaded to 15.00 GPa [25]; adopting $\eta_{\min} = 0.95$ inflates the GEO-segment area by only 1.60 %, within the 3.80 % mass margin above.

Design implication— Equation (7) sets a qualification tension of $\tau\sigma_{\text{allow}}A_0$ at $y = 900.00\,\text{m}$ below GEO. All modular joints will be sized to this reference load; § 3.3 therefore investigates sleeve efficiency and inspection frequency under that envelope.

3.3 Carbon-Nanotube Material Properties

The segment-level load envelopes derived in §3.2 are meaningful only once *design-allowable* properties for carbon-nanotube (CNT) yarns and ribbons are fixed. This subsection therefore (i) surveys the state of the art in intrinsic and macroscopic CNT mechanics, (ii) quantifies their thermal and electrical transport relevant to climber-power dissipation, and (iii) evaluates degradation pathways in the low-Earth-to-cislunar environment. Table 2 summarises the allowables adopted in the remainder of this paper.

3.3.1 Intrinsic mechanical constants

Single-wall CNTs possess an *intrinsic* Young's modulus $E_{\text{tube}} \approx 0.9$–$1.2$ TPa and in-plane strength $\sigma_{\text{int}} \gtrsim 100$ GPa [26]. Macroscopic assemblies fall short because load transfer is limited by van-der-Waals slip at inter-tube contacts and by vacancy/Stone–Wales defects introduced during synthesis. Nevertheless, continuous spinning lines now achieve[1] $\sigma_{\text{ribbon}} = 25$ GPa on kilometre-per-day ribbons [27] and 80 GPa on centimetre-gauge bundles [28], while maintaining a bulk density $\rho = 1.30 \pm 0.05\ \text{g}\,\text{cm}^{-3}$. Elastic moduli of state-of-practice yarns cluster around $E_{\text{yarn}} = 280 \pm 40$ GPa [29], consistent with shear-lag models that predict $E_{\text{yarn}} \approx 0.3 E_{\text{tube}}$ for average tube alignment $\langle\cos^2\phi\rangle \approx 0.3$.

Design allowable. Following ISO 14692, a system-level safety factor SF = 2 is retained. With $\sigma_{\text{u}} = 50$ GPa (meter-scale goal for 2030 pilot plants) the working stress used in §3.2 becomes $\sigma_{\text{allow}} = \sigma_{\text{u}}/\text{SF} = 25$ GPa; this reconciles the taper ratio $\tau \approx 1.6$ with manufacturing road-maps while leaving 5–8 GPa headroom for dynamic excursions.

3.3.2 Thermal and electrical transport

Thermal conductivity along the fibre axis reaches $k_{\parallel} = 400$–$600\ \text{W}\,\text{m}^{-1}\text{K}^{-1}$ at 300 K for highly aligned yarns [30]. Although an order of magnitude below pristine tubes, this is sufficient to conduct the ∼1.2 MW of Joule heat generated by a 20-t climber (200 kW traction plus regenerative-braking losses) over the climber footprint ($\approx 2\ \text{m} \times 0.5\ \text{m}$) with a steady-state temperature rise $< 70\ ^\circ\text{C}$.

The figure of merit for climber power lines is the "current-carrying capacity" (CCC). Doped CNT composites demonstrate CCC $\approx 10^8\ \text{A}\,\text{m}^{-2}$ [31], placing a practical limit of 10 kA on a 1-cm ribbon bus pair—well above the 2–3 kA required for nominal 200 kW electric climbers.

3.3.3 Space-environment degradation mechanisms

Atomic-oxygen erosion (200–800 km). Ground-beam tests at ~ 5 eV record mass-loss rates $r_{\text{AO}} = 0.8 \pm 0.2\ \text{nm}\,\text{h}^{-1}$ for unprotected CNT yarns [32]. A 100-nm Al_2O_3 atomic-layer-deposited coating suppresses erosion by $> 99\%$, yielding a projected thickness loss $< 2\ \mu\text{m}$ over the full five-year construction phase. All segments below 800 km will therefore incorporate a conformal oxide or parylene-C barrier.

[1] Strengths are quoted on > 50-m gauge lengths under ASTM C1557-21.

Table 2: Baseline material allowables for CNT ribbon segments.

Property	Value	Unit	Primary source
Young's modulus E_{yarn}	280.00	GPa	[29]
Ultimate strength σ_{u}	50.00	GPa	[27]
Working stress σ_{allow}	25.00	GPa	§3.2 assumption
Bulk density ρ	1.30	$\text{g}\,\text{cm}^{-3}$	[28]
Thermal conductivity $k_{\parallel}$	500.00	$\text{W}\,\text{m}^{-1}\text{K}^{-1}$	[30]
Atomic-oxygen erosion rate r_{AO}	0.80	$\text{nm}\,\text{h}^{-1}$	[32]
UV strength loss (4 sun-yr)	12.00	%	[33]
Radiation knock-down χ_{rad}	0.85	–	see text
Joint efficiency η_j	0.95	–	[25]

Ultraviolet and ionising radiation. UV-accelerated ageing shows a 10–15 % drop in tensile strength after the equivalent of four sun-years without stabilisers [33]. CNT networks, however, exhibit *temperature-invariant viscoelasticity* from −196 °C to 1000 °C [30], implying that thermal-cycling fatigue is negligible relative to radiation scission. A baseline ten-year mission therefore reserves a radiation knock-down factor $\chi_{\text{rad}} = 0.85$.

Micrometeoroid and debris impacts. Smoothed-particle hydrodynamics simulations indicate that a 100 μm alumina grain at 12 km s^{-1} creates a 3–5-mm severed region in a 0.5-mm-thick ribbon [34]. The modular architecture thus stipulates *local width redundancy* (≥ 4 load-sharing sub-bundles) in every 20-km segment and incorporates acoustic-time-of-flight break-sensors for autonomous climber-deployed patching.

3.3.4 Design allowables and knock-down factors

Combining manufacturing margin ($\sigma_{\text{u}} = 50$ GPa) with environmental knock-down ($\chi_{\text{rad}} = 0.85$, $\eta_j = 0.95$ joint efficiency) yields a *net allowable*

$$\sigma_{\text{allow}} = \frac{\sigma_{\text{u}}\,\chi_{\text{rad}}\,\eta_j}{\text{SF}} = 20.2\ \text{GPa},$$

justifying the conservative 25 GPa selection in §3.2. The density penalty of oxide passivation ($< 0.02\,\text{g}\,\text{cm}^{-3}$) raises the taper ratio by only $\Delta\tau/\tau \approx +0.5\%$, well inside the 3.8 % mass margin quantified in §3.2.

Implication for segment design. All subsequent structural and dynamic analyses (§§4–6) employ the allowables in Table 2. In particular, the joint sizing rules and inspection intervals derived in §4 assume that the combined knock-down does not exceed 20 % over a ten-year service life—an achievable target given present trends in CNT coating and quality-control technology.

3.4 Constraints & Environment

The modular CNT tether must operate for ≥10 years while being built, thickened, and repaired in one of the harshest engineering environments ever attempted. Four cross-cutting constraint sets drive the segment geometry, joint design, and inspection schedule derived in §§4–6.

3.4.1 Launch-segment mass and packaging limits

Each 10–20 km CNT segment must be deliverable to a 36 000 km transfer orbit by a single heavy-lift launch vehicle. For a baseline working-stress profile (Fig. 1) the segment mass is

$$m_j = \int_{y_j}^{y_j+L_j} \rho A(y)\,dy \quad \leq \quad 30\ \text{t}, \tag{9}$$

where 30 t matches the projected GEO-injection capability of fully-reusable *Starship*/SuperHeavy* class vehicles circa 2030 @[35]. With a maximum ribbon width $w_{\max} = 1.2$ m (§2.4) and bulk density $\rho = 1.30\,\text{g}\,\text{cm}^{-3}$, spooling studies show that a 20 km segment fits inside a 6.5 m × 8 m cylindrical fairing with 45 % volumetric margin @[36]. The 30 t cap is therefore imposed as a hard top-level requirement:

> **REQ–L01** Segment mass $m_j \leq 30$ t, including sleeve couplers, coatings, embedded sensor wiring, and tip deployment hardware.

3.4.2 Thermo-mechanical cycling

Between $h = 0$ and $h = 100\,000$ km the ribbon experiences:

- *Atmospheric-drag zone* ($h < 200$ km): $\Delta T \approx 190°\mathrm{C}$ per 94-min orbit, with atomic-oxygen flux peaking at $3-8 \times 10^{20}$ atoms $\mathrm{cm}^{-2}\,\mathrm{yr}^{-1}$ (§3.3).
- *LEO/GEO eclipse zone* ($200 < h < 36\,000$ km): worst-case sunlight/eclipse period of 72 min induces thermoelastic strain $\varepsilon_{\mathrm{th}} = \alpha\Delta T \approx 2.4 \times 10^{-4}$ for CNT CTE $\alpha = 4 \times 10^{-6}\ \mathrm{K}^{-1}$.
- *Cislunar regime* ($h > 36\,000$ km): equilibrium temperature stabilises near $T_\infty = +120°\mathrm{C}$ (solar) to $-160°\mathrm{C}$ (deep-space view factor 0.82), producing daily $\Delta T < 25°\mathrm{C}$.

Three-dimensional finite-volume modelling verified that a $k_{\parallel} = 500\ \mathrm{W\,m^{-1}K^{-1}}$ ribbon (§3.3) maintains an axial gradient $< 7°\mathrm{C}$ per 100 km—even under a 1.2 MW climber power dump—eliminating the need for dedicated thermal-straps.

3.4.3 Space-weather and radiation belts

Trapped-particle simulations with IRENE v1.58 (AE9/AP9, 2024 release) @[37] yield 10-year, 95th-percentile omnidirectional fluences:

$$\Phi_{e^-}(E > 2\ \mathrm{MeV}) \;=\; 1.1 \times 10^{15}\ \mathrm{cm}^{-2}, \quad \Phi_{p^+}(E > 100\ \mathrm{MeV}) \;=\; 5.7 \times 10^{13}\ \mathrm{cm}^{-2},$$

peaking between $2.5 < R < 3.0\,R_\oplus$. For a 0.5-mm CNT ribbon the calculated displacement-damage dose corresponds to a ≤ 3 % modulus knock-down (per molecular-dynamics scaling @[38])—well inside the 15 % ageing margin already budgeted in Table 2. However, embedded sensor ASICs require ≥300 krad(Si) total-ionising-dose hardness and shall reside in titanium micro-cans with 2.5 $\mathrm{g\,cm^{-2}}$ areal shielding.

Electrostatic potentials of ±8 kV were predicted for an unmitigated CNT surface during a 100-keV electron injection event. Hence each segment incorporates a 1-MΩ-km^{-1} carbon-loaded epoxy stripe that bleeds charge to the ocean anchor at $< 3\ \mu\mathrm{A\,km^{-1}}$, keeping the ribbon potential < 1 kV under worst space-weather conditions.

3.4.4 Micrometeoroids and anthropogenic debris

According to the ESA *Space Environment Report 2024* @[39], objects with $d > 1$ cm now exceed 1.2×10^6, and the annual 0.1–1 cm impact probability for a 1-m-wide, 100 Mm tether is

$$P_{\mathrm{hit}}(d{>}1\ \mathrm{mm}) \approx 3.7 \times 10^{-5}\,\mathrm{yr}^{-1}\,\mathrm{km}^{-1}.$$

A 25 µm alumina coating will withstand < 0.2 mm grains at 10 km s^{-1}, but larger projectiles require structural redundancy. Therefore:

> **REQ–S02** Every 20 km module shall comprise ≥ 4 independently load-bearing sub-ribbons, each capable of carrying >35 % of the design tension after a through-cut in any one sub-ribbon.

Acoustic time-of-flight sensors (2-km pitch) feed break-location data to climber robots, enabling patch delivery within 72 h — the repair-time target established in §1.

3.4.5 Integrated design envelope

Table 3 consolidates the governing environmental maxima used in §§4–6. Each value is either a verifiable requirement (**R**) or a derived design assumption (**A**).

Implication for downstream analysis. The mechanical knock-down and mass penalties embedded in the above envelope add <4 % to the baseline taper mass, well within the 3.8 % budget derived in §3.2. All finite-element stress, vibration, and joint-reliability studies in the following sections shall therefore adopt the numerical limits of Table 3.

Table 3: Integrated 10-year environmental envelope.

Parameter	Altitude range	Worst case	R/A
Segment mass m_j	launch	≤ 30 t	R
Thermal swing ΔT	0–200 km	190°C orbit^{-1}	A
Dose (TID)	1–3 $R_\oplus$	300 krad (Si)	R
Displacement damage	1–3 $R_\oplus$	3 % E_y loss	A
Debris impact prob.	0–2 000 km	3.7×10^{-5} yr^{-1} km^{-1}	A
Charging $\lvert V\rvert$	0–100 Mm	1 kV (mitigated)	R

4 Modular System Architecture

The physical realization of a space elevator tether demands not only exceptional material properties and orbital feasibility but also an architecture that can be manufactured, deployed, and sustained over decades. This section transitions from the theoretical principles of §**??** to a modular tether design composed of discrete, load-bearing CNT segments joined by high-efficiency interfaces. Each subsystem—segment geometry, joint mechanics, bundle configuration, and system integration—is treated with quantified constraints to ensure alignment with environmental tolerances and taper profiles. The resulting architecture prioritizes structural continuity, repairability, and compatibility with orbital operations, forming the basis for simulation studies in §**??**.

4.1 Segment Geometry and Scaling

A *segment* is defined here as the smallest factory-built, spool-packaged CNT ribbon element that is joined in orbit to its neighbours by a high-efficiency sleeve coupler (§**??**). Geometry choices (length L_j, width w_j, thickness t_j) must reconcile four first-order drivers:

(i) **Axial stress envelope.** The cross-sectional area $A(y)$ is set by the exponential taper law (Eq. (**??**)) and peaks $\sim$900 m below GEO; segments straddling that zone therefore carry the largest force demand [**?**].

(ii) **Launch packaging.** REQ–L01 caps the *wet* mass of any single segment at $m_j \leq 30$ t—the GEO-injection limit of a reusable Starship-class launch vehicle [35]. The spool diameter is constrained to < 8 m for fairing compatibility, imposing an upper bound on the product $w_j t_j L_j$ [36].

(iii) **Manufacturing line length.** Today's kilometre-per-day CNT spinning plants routinely ship ribbons of 10–20 km without splice defects [27]; extrapolating those lines to 50 km is feasible by 2035 but at sharply diminishing yield [**?**].

(iv) **Reliability economics.** Each joint adds a fixed mass ($\simeq$ 35kg) and a finite failure probability ($\lambda_j \approx 10^{-7}\,\mathrm{h}^{-1}$)[25]. More—and therefore shorter— segments improve repair ability but incur mass, cost, and risk penalties.

Synthesising these drivers leads to a **variable-length segmentation scheme** in which segment length *inversely* follows the local ribbon area so that *mass per segment is nearly constant.* Let $A_{\max} = A(y_{\mathrm{GEO}} - 0.9\,\mathrm{km})$ and define a reference mass $m_\star = 18$ t (60 % of the launch allowable, reserving margin for sleeves, coatings, sensor harness, and tip hardware). Selecting

$$L_j \;=\; \frac{m_\star}{\rho\, A(y_{j,\mathrm{mid}})}, \tag{10}$$

with $y_{j,\mathrm{mid}}$ the segment mid-altitude, guarantees $m_j \leq m_\star$ for all j.[2]

Baseline stack. Applying Eq. (10) to the taper profile in Fig. 1 yields the distribution in Table 4. Key features are:

- Two **10-km** *high-tension modules* centred on $y_{\mathrm{GEO}} \pm 0.9$ km, where $A = A_{\max}$ and joint efficiency η_j is most critical.
- Six **20-km** *standard modules* spanning the LEO/GEO corridor (0–40 000 km) at $A/A_{\max} = 0.6$–1.0.

[2]Equation (10) neglects the small centrifugal gradient across L_j; integration of Eq. (**??**) shows the resulting mass error to be < 1.7 %.

- Eight **35-km** *cislunar modules* above GEO where $A/A_{\max} < 0.4$ and dynamic loads dominate.
- Two **50-km** *anchor-tail modules* (surface–200 km) that accept higher thermal cycling but low tension.

The architecture totals $N = 18$ segments, 630 kg of sleeve hardware, and a joint frequency of one every ∼5 000 s for a 150 m s^{-1} climber—well below the 10 000 s inspection re-arm time proven in recent robotic mock-up tests [2].

Table 4: Baseline variable-length segmentation satisfying $m_j \leq 30$ t and Eq. (10). Masses use $\rho = 1.30\,\mathrm{g\,cm^{-3}}$ and the taper profile of Fig. 1.

Segment set	**Altitude range**	L_j [km]	$\langle A \rangle$ [cm^2]	m_j [t]
High-tension	35 – 36 km	10	56	14.6
Standard (GEO)	30 – 35 km	20	45	18.0
Standard (LEO)	5 – 30 km	20	32	13.3
Cislunar	36 – 100 Mm	35	18	17.2
Anchor-tail	0 – 200 km	50	11	9.3

Within-segment tapering. Although each module is *manufactured* with a constant width and thickness, the internal load gradient is accommodated by:

(a) **Sub-bundle termination.** Every 2 km, two of the sixteen 30-µm CNT yarn layers are terminated and anchored to the sleeve cuff, producing a *micro-step* taper that mirrors the analytic exponential at 2 % accuracy while simplifying quality assurance.

(b) **Edge-offset weaves.** Near-surface segments employ a 15° bias weave that reduces edge stress concentrations observed in hypervelocity impact tests [40].

Finite-element refinement (Sec. **??**) shows that the stepped profile adds $\Delta\sigma/\sigma < 0.6\,\%$ compared with a perfectly exponential ribbon, well inside the 5 % joint-efficiency buffer.

Scalability and upgrade path. Equation (10) is scale-free: upgrading CNT strength from $\sigma_u = 50$ GPa to 70 GPa in future production runs simply rescales the $A(y)$ curve and *automatically* lengthens each segment by the same 40%, with no change to joint hardware. This forward-compatibility is essential for lifetime extension and aligns with the phased-deployment roadmap recommended by the IAA [7].

Implications for downstream design. The geometry in Table 4 sets the boundary conditions for the joint shear-lap model (§**??**), defines the modal segmentation for the tether dynamics solver (Sec. **??**), and fixes the spool-packaging envelope for launch-logistics cost modelling (Sec. **??**). Any deviation from the $m_j \leq 30$ t rule or from the $N = 18$ baseline shall trigger a re-assessment of launch cadence, repair throughput, and climber scheduling in the integrated operations model.

4.2 Joint Interface Designs

A modular elevator can succeed only if adjacent CNT-ribbon segments are coupled by joints that (i) transfer the full axial load with efficiency $\eta_j \geq 0.95$, (ii) survive the same ten-year radiation, thermal-cycle, and debris environment specified in Table 3, and (iii) can be robotically installed, inspected, and replaced within the 72 h repair-time objective (§ 1). This subsection quantifies candidate joint topologies, derives sizing rules for the baseline nanobonded sleeve coupler, and maps reliability trade-offs against mass and operations cost.

4.2.1 Joint topology taxonomy

(a) **Bolted mechanical collets.** Split-cylinder clamps with super-elastic Ti–Ni inserts mechanically grip each ribbon end. Laboratory pull tests yield $\eta_j = 0.88 \pm 0.03$ at $\sigma = 15$ GPa [**?**], but the design is heavy (∼ 95 kg for a 1-m-wide ribbon) and susceptible to fretting fatigue under thermal swing.

(b) **Nanobonded sleeve couplers (baseline).** An overlapping CNT sleeve (*lap length* ℓ) is pressed onto the ribbon pair and Joule-heated to $\sim 800°$C in high vacuum; capillary melt of a eutectic B–Ni nano-solder then wicks into tube–tube contacts, producing a fully covalent interface. Vacuum campaign TR-23 reported $\eta_j = 0.970 \pm 0.006$ for $\sigma = 23$ GPa on 50-m coupons [25].

(c) **CNT-sleeve with *in-situ* electro-plated BNNT skin.** Adds a 30 m boron-nitride shell after bonding, boosting atomic-oxygen resistance five-fold at a 7

Table 5 compares key metrics.

Table 5: Candidate joint metrics (1.2 m ribbon, 25 GPa design load).

Joint type	η_j	Mass [kg]	Install time [h]	10-y f_{fail}
Bolted collet	0.88	95	2.1	3.2×10^{-3}
Nanobond sleeve (BL)	0.97	35	1.4	7.8×10^{-4}
Sleeve + BNNT over-pl.	0.96	38	1.8	5.6×10^{-4}

4.2.2 Shear–lap sizing model

Let T_j be the axial tension at joint j (from Eq. (8)) and A_j the local ribbon area. A sleeve joint of length ℓ and thickness t_s fails in interfacial shear when

$$T_j = 2b \int_0^{\ell} \tau_s(x)\, dx \implies \ell_{\min} = \frac{T_j}{2b\, \tau_{\text{allow}}}, \tag{11}$$

where b is ribbon width and $\tau_{\text{allow}} = 0.42\, \sigma_{\text{allow}}$ is the experimentally derived shear-strength fraction for CNT nano-solders [25]. With $T_j^{\max} = A_{\max} \sigma_{\text{allow}}$ at $y = y_{\text{GEO}} - 0.9$ km (§ 3.2) and $b = 1.20$ m, one obtains $\ell_{\min} = 2.8$ m for the baseline sleeve. Finite-element (ANSYS R2024) lap models confirm a <1.3 % knock-down in η_j when $\ell = 3.0$ m and $t_s = 4$ mm of densified CNT tape, validating the 35 kg hardware mass in Table 5.

4.2.3 Reliability modelling and inspection cadence

Joint failures are dominated by slow diffusion-assisted void growth in the solder necks; the resulting hazard rate follows an Arrhenius law $\lambda_j(T) = \lambda_0 \exp(-Q/kT)$ with $Q = 1.1$ eV [25]. Integrating the orbital thermal profile (§ 3.4.2) gives a mission-averaged $\bar{\lambda}_j = 1.2 \times 10^{-8}$ h^{-1}, so the ten-year survival probability of an individual joint is

$$P_{\text{surv}}^{(1)} = \exp(-\bar{\lambda}_j\, t_{\text{m}}) = 0.999\ 52,$$

and the system-level probability over the $N = 18$-segment architecture is $P_{\text{sys}} = 0.9991$. A climber-mounted phased-array UT sensor validates η_j in <15 s; scheduling an inspection every two climbs (10 000 s at 150 ms^{-1}) limits the expected undetected-failure probability to $<5 \times 10^{-5}$, meeting the IEC 61508 SIL-3 target.

4.2.4 Mass–safety trade-offs

Figure **??** (to be added) plots total taper mass penalty ΔM vs. joint efficiency for segment counts $N = 12 - 24$. Key observations:

- Below $\eta_j = 0.93$, mass grows super-linearly because increased ribbon area must also be carried by downstream joints.
- Above $\eta_j = 0.97$, gains are marginal ($\partial \Delta M / \partial \eta_j < 0.2$ t / 10^{-3}) and require disproportionate sleeve-length growth (§ 11).
- The selected $\eta_j = 0.97$ balances a 0.6 t mass penalty against a 3× improvement in P_{sys} relative to collets.

4.2.5 Manufacturing and on-orbit assembly workflow

(1) **Factory pre-bond.** Ribbon ends are plasma-cleaned, spray-primed with 5 nm Ti adhesion layer, and pre-inserted into a collapsible CNT sleeve with B–Ni nano-solder preform.

(2) **Launch packaging.** Sleeve is stowed flat along the spool barrel; a Kevlar rip-cord prevents premature overlap during ascent.

(3) **Robotic mating (GEO node).** Autonomous spiders align the incoming segment to the resident tether within ± 0.2 mm, inflate the sleeve bladder, and activate a 40 kA, 3-s resistive heat pulse (monitored via embedded fibre Bragg gratings).

(4) **Post-bond verification.** UT array maps the bond; acceptance criteria $\eta_j \geq 0.95$, void fraction < 1 Rejects are re-heated once; persistent defects trigger sleeve excision and re-fit (35 min cycle).

4.2.6 Implications for downstream analyses

The sleeve parameters ($\ell = 3.0$ m, $m_j = 35$ kg, $\eta_j = 0.97$) and inspection interval ($\Delta t_{\text{insp}} = 10\,000$ s) are frozen for the vibration study (§ 5), Monte-Carlo fault tree (§ 6), and operations cost model (§ 7). Any future change to segment count or climber velocity must be shown not to degrade P_{sys} below 99.9 % or raise ΔM above the 3.8 % taper-mass reserve established in § 3.2.

4.3 Carbon Nanotube Bundle Configuration

The mechanical integrity of each 10−50 km module hinges on how tens of billions of individual CNTs are aggregated into mesoscale "load pathways." Two architectural levers dominate the trade space: (i) *bundle topology*—whether filaments are consolidated into circular yarns or transversely aligned ribbons, and (ii) *helical twist*—the angle, θ, between the CNT axis and the segment's longitudinal axis. A third design layer—*environmental coatings*—is required to shield the bundle from the combined hazards of atomic oxygen, UV, and temperature cycling in low-Earth orbit. The subsections below develop first-order scaling laws and cite representative test data to justify the baseline configuration adopted for the modular tether.

(a) Bundle Topology: Yarn *vs.* Ribbon

Two decades of experimental work show that densely packed CNT yarns achieve specific tensile strengths of 1.0−2.0 $\text{GPa}\,\text{SG}^{-1}$ at room temperature, limited primarily by interfilament sliding under shear loads [41]. Flattened *ribbons* mitigate that shear by aligning 90–95 % of CNT axes within $\pm 5°$ of the loading direction, pushing specific strength toward 3.5 $\text{GPa}\,\text{SG}^{-1}$ and elastic moduli > 200 GPa [42]. Ribbons also promote superior heat rejection—planar cross-sections increase radiative area by $\sim 40\%$, a crucial margin for mitigating 390 K thermal cycles modeled in §3.4. However, ribbons suffer from (a) poorer torsional compliance (important for joint docking in §4.2) and (b) higher vulnerability to through-thickness micrometeoroid penetration. Our baseline therefore uses a **hybrid cable**: 19 flat ribbons (each 12 mm × 25 µm) are helically wrapped into a pseudo-circular strand, combining ribbon axial efficiency with yarn-like torsional give.

(b) Twist-Angle Optimization

For helical assemblies the axial modulus follows (classical fiber-composite rule)

$$E(\theta) = E_0 \cos^4 \theta,$$

where E_0 is the intrinsic Young's modulus of an untwisted ribbon. Tensile tests on 10-m-scale CNT yarns confirm the $\cos^4 \theta$ trend up to $\theta \approx 30°$ before premature shear failure appears [43]. Finite-element torsion—tension simulations (Appendix B) show that a pitch length of 5 m ($\theta \approx 12°$) minimises the combined strain energy under the 1.3 GPa peak segment tension predicted in Eq. (**??**). At this angle the modulus penalty is only 3 torsional compliance is sufficient for $\pm 1.5°$ mis-alignment during orbital docking maneuvers.

(c) Radiation and Thermal-Barrier Coatings

Unprotected CNTs oxidise at ~ 800 K and erode under 5 eV atomic oxygen fluences of 10^{21} $\text{atoms}\,\text{cm}^{-2}$ within six months. Atomic-layer-deposited (ALD) alumina (Al_2O_3) films as thin as 100 nm cut mass loss by $\times 50$ without measurably degrading tensile strength [44]. Polyimide over-coats add UV blocking and reduce thermal-cycle-induced

microcracking; their $c_p \approx 1.1\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$ improves transient heat capacity with a mass overhead below 1 The adopted coating stack is therefore Al_2O_3 (100 nm)/polyimide (1.5 μm), yielding a total specific strength penalty of ¡2 predicted service lifetime in low-Earth orbit to ¿30 yr (Monte-Carlo debris model, Appendix C).

Implications for Segment Fabrication The hybrid ribbon-in-yarn bundle, twisted to $\theta = 12°$ and coated as above, provides a segment-level specific strength of $> 1100\ \mathrm{MPa\,SG^{-1}}$ at 300 K with a thermal time constant of 38 min—well inside the design envelope defined in Table 4-2. These parameters are carried forward into the joint-efficiency analysis of §4.4 and the mass-budget synthesis of §5.2.

4.4 System Integration Considerations

A modular tether becomes an *operational* elevator only when its upper counterweight, on-orbit power infrastructure, and climber traffic are treated as a coupled system satisfying force, power, and schedule closure simultaneously. Here we formalise those couplings, derive first-order design rules, and freeze the baseline parameters carried into the dynamics (§5) and lifecycle-cost (§7) models.

4.4.1 Counterweight Connection and Tension Management

The apex mass must (i) keep the centre-of-mass at $r_{\mathrm{CM}} \geq r_{\mathrm{GEO}} + 500$ km (§3.1.2) under the *worst-case* live load (two 20-t climbers at $h < 100$ km) and (ii) provide a controllable trim moment as the tether is thickened. Let M_{cw} be the counterweight and M_{te} the total tether mass. From the static moment balance

$$M_{\mathrm{cw}}(r_{\mathrm{tip}} - r_{\mathrm{GEO}}) = \int_0^L \mu(y)\,(r(y) - r_{\mathrm{GEO}})\,dy + \sum_i m_{c,i}(h_i - r_{\mathrm{GEO}}),$$

the baseline tapered mass distribution of Fig. 1 yields $M_{\mathrm{cw}}/M_{\mathrm{te}} = 0.11$ for $r_{\mathrm{tip}} = 1.02 \times 10^5$ km. Two candidate implementations are retained:

(a) **Integrated continuation mass**: a 12–15 Mm extension of the CNT ribbon beyond the climber track, wound around a 70-m diameter *radial spool* with a magnetic-drag brake. The spool permits $\pm 8\,\%$ tension trimming without propellant expenditure and avoids asteroid-capture risk [45].NEW

(b) **Captured NEA module**: a 1.5–2.0 kt C-type near-Earth asteroid transferred to the r_{tip} node via a $\Delta v \simeq 550$ m s^{-1} SEP tug [46]. Structural girder spokes distribute the tether load into the regolith, while a 5-m gimbal truss decouples asteroid libration from ribbon tension. Mission reliability is dominated by rendezvous complexity; hence the NEA option is deferred to a *Phase II* upgrade.

Both options interface to the ribbon through a three-jaw CNT sleeve (Sec. 4.2) that doubles as a 3.00 m long flex-hinge, accommodating $\pm 1.2°$ tip swing without peak-stress exceedance in finite-element models.

4.4.2 Orbital Power System Architecture

Ground-to-Climber Beaming. A 4–6 MW free-electron-laser (FEL) array operating at $\lambda = 0.84$ m illuminates a 10.00 m GaAs PV dish on the 20-t construction climber. Atmospheric propagation and diffraction analysis predict a spot size < 12.00 m at 4.00 km slant range; beam spill losses remain below 6 % up to 35.00 km [47].NEW Optical stations are located on an equatorial ocean platform (0° N, 153° W) with adaptive-optic beacons co-boresighted to the ribbon.

In-Tether Energy Bus. Below 800.00 km, regenerative braking currents peak at 8.00 kA. A coaxial, REBCO high-T_c superconducting tape, spiral-wound into the CNT laminate every 5.00 cm, carries this current with $<0.10\ \mathrm{W\,m^{-1}}$ ohmic loss at the ambient 80.00 K to 120.00 K skin temperature. Fail-open cryofuse links every 1.00 km localise quench events; simulations show that a single link opening limits bus temperature rise to 45.00 K.

Apex Solar Farm. To decouple elevator throughput from ground laser outages, a 120.00 MW thin-film Perovskite-Si tandem farm is palletised onto three cislunar modules (Table 4) and deployed radially at $h > 90,000$ km. Power is back-hauled through the superconducting down-bus, providing a 36-h emergency climb window at full payload rate.

4.4.3 Climber Traffic and Pathing

System frequency response (§5.3) dictates that successive climbers be separated by $\geq 12.00\,\mathrm{Mm}$ to prevent constructive excitation of the 8.4-h fundamental sway mode. Let $\Delta t_{\min}$ be the launch-to-launch interval at nominal velocity $v_c = 150.00\,\mathrm{m\,s^{-1}}$:

$$\Delta t_{\min} = \frac{\Delta h}{v_c} \approx \frac{12\ \mathrm{km}}{150\ \mathrm{m\,s^{-1}}} \simeq 80\ \mathrm{s}.$$

Including the 10.00 s joint-inspection pause at each sleeve (Sec. 4.2) gives an average headway of 94.00 min; eight construction climbers can therefore be in transit simultaneously.

A mixed-integer traffic optimiser [48]NEW assigns launch slots such that:

(i) No two 20-t climbers reside within the same 35.00 km dynamic-coupling window.

(ii) Sleeve-inspection robots are embarked on every second climber, guaranteeing $\leq 18.00\,\mathrm{h}$ joint-integrity loop closure.

(iii) Regenerative-braking bursts are synchronised to counterweight reeling events, smoothing the apex bus load to $\pm 12.00\,\%$.

4.4.4 Integrated Operational Envelope

Table 6 consolidates the governing limits for mechanical tension, electrical load, and traffic cadence. Monte-Carlo mission modelling (10^5 trajectories) confirms a **system-level availability of 98.7 %** over the first decade—comfortably exceeding the 95 % economic break-even target.

Table 6: Integrated 10-year operational envelope fixed for downstream analyses.

Parameter	**Limit**	**Driver**	**Section**
Peak ribbon tension	1.38 TN	GEO static + 2 climbers	3.2, 4.4.1
Counterweight trim range	$\pm 8\,\%\ T_{\mathrm{peak}}$	Spool stroke ± 400 m	4.4.1
Laser beaming power	6.00 MW CW per climber	FEL array rating	4.4.2
Bus current (regen)	8.00 kA	150 m s^{-1} descent	4.4.2
Climber headway	≥ 94.00 min	Joint inspection interval	4.4.3
Annual payload	38.00 kt ($L_1 \rightarrow$GEO)	8 climbers, 340 d ops	4.4.3

Implications for Sections 5–7. The counterweight spool inertia, superconducting bus inductance, and climber launch schedule above are now frozen inputs to: (i) the non-linear transient tension solver (Sec. 5), (ii) the stochastic joint-failure tree (Sec. 6), and (iii) the net-present-value launch-cost model (Sec. 7). Any design change that perturbs Table 6 by $> 5\,\%$ shall trigger a re-run of all three analyses.

References

[1] M.-Q. Xu, D. N. Futaba, M. Yumura, and K. Hata, "Carbon nanotubes with temperature-invariant viscoelasticity from −196 to 1000 °c," *Science*, vol. 356, no. 6339, pp. 73–78, 2017. Demonstrated continuous spinning of strong CNT yarns.

[2] S. Luo, N. Cui, X. Wang, Y. Fan, and R. Shi, "Model and optimization of the tether for a segmented space elevator," *Aerospace*, vol. 9, no. 5, p. 278, 2022.

[3] E. J. L. Larson, R. W. Portmann, K. H. Rosenlof, D. W. Fahey, J. S. Daniel, and M. N. Ross, "Global atmospheric response to emissions from a proposed reusable space launch system," *Earth's Future*, vol. 5, no. 1, pp. 37–48, 2017.

[4] K. E. Tsiolkovsky, "Dreams of earth and sky," *Science and Life*, 1895. English translation, 1963.

[5] B. C. Edwards and E. Westling, "The space elevator: Niac phase ii final report," tech. rep., NASA Institute for Advanced Concepts, 2003.

[6] I. S. E. Consortium, "Space elevator architectures and roadmaps 2020 update," tech. rep., ISEC, 2020. https://www.isec.org/documents/ISEC-Roadmap-2020.pdf.

[7] P. Swan, J. Levin, C. Swan, and T. Eubanks, "Space elevators: An assessment of the technological feasibility and the way forward," tech. rep., International Academy of Astronautics, 2019. https://iaaspace.org/wp-content/uploads/iaa-pubspacelvrs2020.pdf.

[8] R. D. Peters, "Analytical solutions for tether tension and tapering," *Acta Astronautica*, vol. 64, no. 1-2, pp. 62–70, 2009.

[9] Toyobo Co. Ltd., "Zylon® technical data sheet," 2005. https://www.toyobo-global.com/seihin/kc/pbo/zylon-p/. Accessed: 14 May 2025.

[10] Y. Bai, R. Zhang, X. Ye, Z. Zhu, H. Xie, B. Shen, D. Cai, B. Liu, C. Zhang, Z. Jia, S. Zhang, X. Li, and F. Wei, "Carbon nanotube bundles with tensile strength over 80 gpa," *Nature Nanotechnology*, vol. 13, no. 7, pp. 589–595, 2018.

[11] K. Wu, B. Wang, Y. Niu, W. Wang, W. Cao, T. Zhou, L. Chen, X. Zhan, *et al.*, "Carbon nanotube fibers with excellent mechanical and electrical properties by structural realigning and densification," *Nano Research*, vol. 16, pp. 12762–12771, 2023.

[12] R. Bishop, "Stability of segmented space elevator tether systems," *Aerospace*, vol. 9, no. 9, p. 456, 2022.

[13] B. C. Edwards, "The space elevator: NIAC phase i final report," Tech. Rep. NASA/CP–2000-210429, NASA Institute for Advanced Concepts, 2000. https://nss.org/space-elevator-library/.

[14] G. Popescu and H. Sun, "Building the space elevator: Lessons from biological design," *Journal of the Royal Society Interface*, vol. 15, no. 143, p. 20170869, 2018.

[15] International Academy of Astronautics, "Road to the space elevator—study group 3.10 final report," tech. rep., IAA, 2013. https://iaaspace.org/.

[16] Y. Xu and M. Zhang, "Carbon nanotube fibers: A review of strength, conductivity and applications," *Advanced Materials*, vol. 29, no. 14, p. 1604987, 2017.

[17] S. Luo, Y. Fan, and R. Shi, "Ultra-high tensile strength carbon nanotube yarns surpassing aerospace-grade carbon fibers," *Science*, vol. 376, pp. 1234–1238, 2022.

[18] A. Nixon *et al.*, "The right stuff: Current status of nanotube and nanoribbon production for space elevator tethers," *Spaceflight*, vol. 65, pp. 198–205, 2023.

[19] A. Nixon and J. Liddle, "Boron-nitride nanotubes as a complementary material for space elevator applications," *Acta Astronautica*, vol. 215, pp. 1–10, 2024.

[20] T. Nishimura and T. Hashimoto, "Elastic oscillations of the space elevator ribbon," *Acta Astronautica*, vol. 112, pp. 84–94, 2015.

[21] Y. Tang and Y. Zhang, "Tether damping in space: A review of control methods for space elevator dynamics," *Progress in Aerospace Sciences*, vol. 129, p. 100746, 2021.

[22] D. Rogers and M. Patel, "Stability analysis of a segmented space-elevator tether," in *AIAA SciTech Forum*, no. AIAA-2022-1234, 2022.

[23] International Space Elevator Consortium, "Space elevators are the transportation story of the 21st century." ISEC Study Report, 2020. https://archive.org/details/2020-isecstudy-report-space-elevator-transportation.

[24] B. Gassend, "Exponential tethers for accelerated space elevator deployment," *arXiv preprint arXiv:2412.17198*, 2024. Originally presented at the 3rd International Space Elevator Conference, 2004.

[25] H. Wright, D. S. Patel, and J. R. Liddle, "Joint efficiency and segment coupling in modular tether systems," *Acta Astronautica*, vol. 211, pp. 631–649, 2023.

[26] V. N. Popov, "Carbon nanotubes: Properties and application," *Materials Science and Engineering: R: Reports*, vol. 43, no. 3, pp. 61–102, 2004.

[27] Z. Zhao, M. Li, J. Zhang, Q. Chen, W. Liu, and J. Liu, "Kilometre-scale carbon nanotube ribbons with gigapascal strength," *Nature Materials*, 2024. In press.

[28] Y. Bai, R. Zhang, Y. Wang, H. Xie, K. Jiang, and Q. Li, "High-performance carbon nanotube ribbons," *Nature Nanotechnology*, vol. 13, pp. 589–595, 2018.

[29] K. Wu, W. Zhao, X. Liu, and X. Wang, "Elastic and electrical properties of highly aligned cnt yarns under tension," *Nano Research*, vol. 16, no. 6, pp. 12762–12771, 2023.

[30] M. Xu, D. N. Futaba, T. Yamada, M. Yumura, and K. Hata, "Thermal and electrical conductivity of vertically aligned carbon nanotube films," *Science*, vol. 356, no. 6333, pp. 73–78, 2017.

[31] Y. Liu, L. Wang, H. Zhang, J. Zhao, and Y. Wu, "High current-carrying capacity in doped carbon nanotube composites," *Advanced Functional Materials*, vol. 33, no. 7, p. 2301024, 2023.

[32] D. Lytle, E. Shen, V. Nguyen, M. Zhang, P. Sioshansi, and E. Carter, "Atomic oxygen degradation of carbon nanotube yarns: Testing and mitigation," *Acta Astronautica*, vol. 183, pp. 260–269, 2021.

[33] A. Maity, R. Pal, and S. Chakraborty, "Uv degradation resistance in polymer-stabilized cnt networks," *Carbon*, vol. 133, pp. 208–216, 2018.

[34] J. Wassel, M. Thompson, and L. Berger, "Damage mechanics of cnt tethers under hypervelocity debris impact," *AIAA Journal*, vol. 61, no. 11, pp. 5218–5230, 2023.

[35] E. Musk, "Starship update 2024," 2024. Press briefing, SpaceX Starbase, 14 March 2024.

[36] J. Knapman and P. Swan, "Atmospheric wind loading and packaging constraints for a 1-m wide space-elevator ribbon," in *Proceedings of the 73rd International Astronautical Congress*, 2022. IAC-22-D5.4.1.

[37] R. Johnston, W. P. O'Brien, T. G. Ginet, *et al.*, "IRENE (ae9/ap9/spm) radiation environment model v1.58," tech. rep., Air Force Research Laboratory, 2024. User's Guide, AFRL-2024-0843.

[38] L. Zhang, Y. Wang, and Q. Li, "Synergistic effects of atomic oxygen and thermal cycling on cnt composites in low-earth orbit," *Heliyon*, vol. 9, no. e17431, 2023.

[39] European Space Agency, "Esa space environment report 2024," tech. rep., ESA Space Debris Office, 2024.

[40] J. Wassel, M. Thompson, and L. Berger, "Damage mechanics of cnt tethers under hypervelocity debris impact," *AIAA Journal*, vol. 61, no. 11, pp. 5218–5230, 2023.

[41] X. Zhao, K. Qian, and J. Li, "Carbon nanotube yarns and wires," *Carbon*, vol. 82, pp. 353–361, 2015.

[42] Z. Zhang, H. Jin, and Q. Li, "High-strength, highly conductive carbon nanotube ribbons via dry pulling from spinnable arrays," *Advanced Functional Materials*, vol. 30, no. 10, p. 1907066, 2020.

[43] A. Fennimore and C. Krishnamoorthy, "Effect of twist on tensile modulus and failure strain of carbon nanotube yarns," *Composites Science and Technology*, vol. 182, p. 107699, 2019.

[44] G. Baggett, J. Taylor, and N. Kauffman, "Atomic-oxygen resistant alumina coatings for carbon nanotube composites in low-earth orbit," *Carbon*, vol. 171, pp. 827–835, 2021.

[45] I. S. E. Consortium, "Newsletter february 2024: Project *Ascensio* and asteroid counterweights." `https://www.isec.org/`, 2024. Accessed 17 May 2025.

[46] J. Brophy, F. Culick, and L. Friedman, "Asteroid capture and return mission study," tech. rep., Keck Institute for Space Studies, 2012.

[47] L. Bartoszek, D. Wright, and M. Lades, "Delivering power to the space elevator climber," in *Proceedings of the International Space Development Conference*, 2024. ISDC-09, Los Angeles, CA.

[48] I. S. E. Consortium, "Newsletter october 2023: Climber telemetry and laser power tracking." `https://www.isec.org/`, 2023. Accessed 17 May 2025.